

III. Plant Hormones

Notes

- The **plant hormones** (also known as phytohormones) are such organic substances which are formed in one tissue or organ of the plants and are then transported to other sites where they produce specific effects on growth and development.
- Plant hormones are single molecules, produced within plants, that occur in extremely low concentrations. Unlike in animals (in which hormone production is restricted to specialized glands), each plant cell is capable of producing hormones.
- The plant hormones are either **growth promoter** or **growth inhibitor**.

1. Growth Promoters :

- These plant hormones promotes the growth of plants. Under this category three hormones have been placed-
 - (i) Auxins
 - (ii) Gibberellins
 - (iii) Cytokinins
- (i) **Auxins :**
 - Auxins are one of the most important groups of plant hormones because of their many-sided role in plants. These substances were also the first growth factors identified as plant hormones.
 - The chief natural auxin is **indole-3-acetic acid** (IAA- $C_{10}H_9O_2N$). It is the most common auxin found in plants.
 - Other naturally occurring auxins are indole-3-acetaldehyde, indole-3-pyruvic acid, indole-3-acetonitrile, indole-3-ethanol.
 - In addition to these auxins, certain synthetic chemicals compounds have been found to act as auxins. These are-
 - Indole-3-butyric acid
 - 1-Naphthaleneacetic acid-NAA
 - Phenylacetic acid
 - 2, 4-D (dichlorophenoxyacetic acid)
 - 2, 4, 5-T (Trichlorophenoxyacetic acid)

Functions of Auxins :

- The primary physiological effect of auxin on the growth of a plant is by its effect on the prolongation of cells in shoot.
- Auxins control the tropic movements as **phototropism** and **geotropism** in plants.
- Auxins enhance the **apical growth** and retard the growth of axillary buds.
- Auxins promote the **differentiation** of tissues and plant organs; for example-root development in grafting by applying auxins (root initiation).

- Auxins are used to produce parthenocarpic fruits (seedless fruits).
- Auxins retard the abscission of fruits and leaves.
- 2-4D and 2,4,5-T are used as **weedicides** (to eradicate weeds).
- Auxins retard the growth of buds. On applying auxins in cold storage potato is stored for a long duration.

(ii) Gibberellins :

- The history of the discovery of gibberellins dates back to the 19th century when the Japanese farmer noticed that certain diseased rice plants grew abnormally thin and tall. They called it '**Bakaene**' or **foolish seedling** disease because it made the young rice plants grow ridiculously tall. Infection by a fungus – **Gibberella fujikuroi** was responsible for the disease.
- A plant hormone secreted by the above-mentioned fungus was responsible for the foolish seedling disease, which was given the name gibberellin.
- It was later discovered that gibberellins are also produced by the plants themselves and they control multiple aspects of development across the life cycle.

Functions of gibberellins :

- The gibberellins have been shown to be effective in protecting the apical meristem from the inhibitory effect of the endogenous growth inhibitors such as **dormin**.
- Gibberellins promote shoot growth mainly by accelerating the roles of **cell elongation** and cell division in the sub-apical meristem region where young internodes are developing. **Lang** (1960) demarcated marked increased in gibberellin level at the time of bolting when a **rosette** growth form (dwarf) starts growing in a longitudinal direction.
- Gibberellins do not affect the growth of roots.
- Other functions of gibberellins are flowering, germination of seeds, parthenocarpy, breaking dormancy, cell division in cambial zone, change in shape and size of leaves and activity of enzymes.
- It is used as the spraying agent to increase sugarcane yield by lengthening of the stem.

(iii) Cytokinins :

- Cytokinins are such organic substances which promote the **cytokinesis** during cell division.
- The **kinetin** and **zeatin** act as cytokinins.
- Coconut milk has cytokinin.
- Cytokinins control cell division, morphogenesis, counteraction of apical meristem, breaking the dormancy, delay of senescence (chlorophyll disappearance and protein degradation).
- They are used to keep flowers fresh for a longer time.

2. Growth Inhibitors :

- These hormones inhibit the growth of plants. Under this category, two hormones are placed –
(i) Absciscic acid
(ii) Ethylene

(i) Absciscic acid (ABA) :

- The physiological roles of ABA are as follows :
- Regulation of dormancy of buds and seeds.
- Acceleration of senescence of leaves.
- Inhibits plant growth, metabolism and germination of seeds.
- Inhibits gibberellin-stimulated growth. Hence it is also known as **anti-gibberellin**.
- Inhibits gibberellin induced α -amylase formation in barley aleurone.
- Promotes ageing and abscission of leaves.
- It is widely used as a spraying agent on trees to regulate dropping of fruits.

(ii) Ethylene ($\text{CH}_2 = \text{CH}_2$) :

- Ethylene is a gaseous plant hormone.
- Ethylene is produced in minute quantity by plant tissues but is active at extremely low concentrations.
- It is synthesized in plant from the **methionine**, an essential amino acid produced in the chloroplast.
- Ethylene has several effects on the vegetative and reproductive growth of plants. It is used as both plant growth promoter and plant growth inhibitor.
- The effect of ethylene are as follows :
- Enhances respiration rate through the **ripening of fruits**. This phenomenon is called as 'respiratory climacteric'.
- Induction of epinasty.
- Inhibition of elongation growth in stems and roots.
- Stimulation of cells to grow isodiametrically rather than longitudinally thus enhancing radial growth rather than elongation growth in stems and roots.
- Induction of root hair formation.
- Promotion of leaf and flower abscission and senescence.
- It is applied to rubber trees to stimulate the flow of latex.

Question Bank

1. Which one of the following is not a plant hormone ?

- (a) Cytokinin (b) Ethylene
(c) Insulin (d) Gibberellin

U.P.P.C.S. (Mains) 2011

Ans. (c)

Insulin is a hormone produced in the pancreas by islets of Langerhans, which regulates the amount of glucose in the blood. The lack of insulin causes diabetes. Cytokinin, ethylene and gibberellin are plant hormones.

2. Which one of the following is a plant hormone?

- (a) Insulin (b) Thyroxine
(c) Estrogen (d) Cytokinin

U.P. R.O./A.R.O. (Mains) 2021

U.P.P.C.S. (Pre) (Re. Exam) 2015

U.P.P.C.S. (Mains) 2016

Ans. (d)

Plant hormones are naturally occurring special organic compounds found in plants. They affect and control many metabolic activities after being transported to different parts of the plants in very low quantity. Auxins, gibberellin, cytokinin, absciscic acid, and ethylene are examples of the plant hormone.

3. Which of the following is a plant hormone?

- (a) Insulin (b) Thyroxine
(c) Estrogen (d) Cytokinin
(e) None of the above

Chhattisgarh P.C.S. (Pre) 2014

Ans. (d)

See the explanation of above question.

4. Which one of the following is a plant hormone?

- (a) Thyroxine (b) Cytokinin
(c) Insulin (d) None of the above

U.P. Lower Sub. (Pre) 2015

Ans. (b)

See the explanation of above question.

5. Match List-I with List-II and select the correct answer from the code given below the lists.

List-I (Plant hormone)	List-II (Function)
A. Auxin	1. Cell elongation
B. Gibberellin	2. Cell division
C. Cytokinin	3. Growth inhibition
D. Absciscic acid	4. Apical dominance

Code :

A	B	C	D
(a) 2	3	4	1
(b) 4	1	2	3
(c) 3	4	2	1
(d) 4	1	3	2

U.P. R.O./A.R.O. (Mains) 2016

Ans. (b)

The correctly matched lists of plant hormones and their functions are as follows :

List-I (Plant hormone)	List-II (Function)
Auxin	– Apical dominance
Gibberellin	– Cell elongation
Cytokinin	– Cell division
Absciscic acid	– Growth inhibition